



DOUGLAS COLLEGE

CSIS 4495-002 (Applied Research Project)

Progress Report 5

Project name: **EcoTots**

Submitted To:

Prof. Padmapriya Arasanipalai Kandhadai

Submitted By:

Gurkanwal Singh

ID: 300375339

Work Log Table

Date	Number of Hours	Description of Work done
April 2	3	Improved CSS styling, Worked on Chatbot feature.
April 3	3	Implemented Database connection with Chatbot.
April 6	3	Completed connection with Chatbot.
April 6	1	Worked on Progress Report 5.

Description of work done:

In this phase, I continued enhancing the chatbot integration by completing the connection between the frontend UI and the backend chatbot logic. I resolved communication issues between the chatbot and the database layer, ensuring that API requests were correctly routed and processed.

Additionally, I improved the overall CSS styling of the project, focusing on visual consistency and responsiveness across different devices. These enhancements contributed to a more seamless and user-friendly experience.

I also contributed to the ongoing development of the chatbot's database integration, further refining the structure and planning of MongoDB collections and data flow, although the full connection is still in progress. Finally, I compiled and documented these updates in **Progress Report 5**, detailing completed tasks, design improvements, and the current status of chatbot-database connectivity.

I faced few problems while doing my tasks, they are as follows:

Ollama Integration Difficulties: Encountered continued challenges in fully integrating Ollama with the backend, particularly in configuring its environment and ensuring compatibility with existing services. Additional time was required to troubleshoot setup issues. It was the most challenging step in completing the project, as I was new to this concept, and implementing it quickly after learning made it quite a hassle.

Backend and Database Connectivity: Faced problems establishing a stable connection between the chatbot and MongoDB. Query structuring and asynchronous handling needed extra attention to avoid request failures and ensure accurate data flow.

CSS & UI Refinements: While enhancing the styling, maintaining consistency across components proved time-consuming, especially while aligning visual updates with dynamic chatbot interactions.

Repo Check-in

The files/folders I have checked into the repo are as follows:

- Improved CSS, Structured the project:
 - Implementation/ecoTots/api/index.js
 - Implementation/ecoTots/client/src/pages/About.jsx
 - Implementation/ecoTots/client/src/pages/CreateListing.jsx
 - Implementation/ecoTots/client/src/pages/Home.jsx

- Chatbot Completion:
 - Implementation/ecoTots/api/controllers/chatbot.controller.js
 - Implementation/ecoTots/api/index.js
 - Implementation/ecoTots/api/models/chatbot.model.js
 - Implementation/ecoTots/api/routes/chatbot.route.js
 - Implementation/ecoTots/client/src/pages/ChatbotPage.jsx